Timing Search[1]

Online Appendix

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A One-Sided Timing Search

The one-sided timing search problem considers an agent who observes market conditions at Poisson rate λ and chooses when to act. With i.i.d. payoffs, the reservation value satisfies the standard McCall equation

$$rv^* = -c + \lambda \int_{v^*}^{\bar{v}} (v - v^*)\, dF(v),$$

and timing search is isomorphic to cross-sectional search.

Under Ornstein–Uhlenbeck payoffs ($dv = \theta(\bar{v} - v)dt + \sigma\, dW$), two results emerge with no cross-sectional analog. First, the reservation value depends on mean-reversion speed θ: $\partial v^*/\partial\theta < 0$ when $v^* > \bar{v}$ and > 0 when $v^* < \bar{v}$. Second, the option value satisfies $\frac{\partial}{\partial v}[W(v) - v] < -1$ for $v \gg \bar{v}$ (agents rush to act at high payoffs), versus exactly -1 under i.i.d. search.

B Matching Microfoundation for Gaussian Profits

This appendix provides a matching-theoretic microfoundation for the Gaussian matching efficiency function $m(n) = \exp(-\frac{\varphi}{2}(n - n_p)^2)$ stated in Assumption 2 of the main text (Gaussian scale efficiency).

Consider a market with n active firms. Each potential match between a firm and a trading partner has a quality θ drawn from a normal distribution $\theta \sim N(n_p, 1/\varphi)$. A firm operating at market thickness n captures matches with quality near n: specifically, matches within distance ε of n on the quality dimension. The density of good matches available to the firm is proportional to the normal density evaluated at n:

$$m(n) \propto \frac{1}{\sqrt{2\pi/\varphi}} \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big).$$

This is the Hotelling–Salop logic applied to market thickness: firms compete for matches along the density dimension, and the normal distribution generates the Gaussian kernel. Absorbing the normalizing constant into π_0 yields $\pi(n) = \pi_0 \exp(-\frac{\varphi}{2}(n - n_p)^2)$.

The parameter φ inherits its interpretation from the variance of match qualities: $\varphi = 1/\mathrm{Var}(\theta)$. Industries with homogeneous match qualities (low variance, high φ)

have a narrow peak and are sensitive to scale deviations. Industries with heterogeneous qualities (high variance, low φ) have a broad peak and are robust to deviations.

C Derivation of the Timing Friction

This appendix derives the exponential-quadratic timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ as the continuous-time limit of proportional adjustment costs.

Discrete-time formulation Consider a discrete-time economy with period length $\Delta > 0$. In each period, effective output is reduced by a fraction proportional to the square of the adjustment speed:

$$y_t = \pi(n_t) \cdot \left[1 - \tfrac{\mu}{2}\left(\frac{n_{t+\Delta} - n_t}{\Delta}\right)^2 \Delta\right].$$

The adjustment cost is proportional to the square of the rate of change $(\Delta n/\Delta)^2$, scaled by the period length Δ. The parameter $\mu > 0$ governs the severity of the friction.

Continuous-time limit As $\Delta \to 0$, write $\dot{n} = \lim_{\Delta\to 0}(n_{t+\Delta} - n_t)/\Delta$ and note that $1 - x \approx e^{-x}$ for small x. The effective output becomes:

$$y(t) = \pi(n(t)) \cdot e^{-\frac{\mu}{2}\dot{n}(t)^2} = \pi(n(t)) \cdot \xi(\dot{n}(t)).$$

Robustness of the quadratic form Any symmetric friction $\xi(\dot{n})$ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$, where $\mu = -\xi''(0)$. The leading-order dynamics—in particular $\omega = \sqrt{\varphi/\mu}$—depend only on μ, not on higher-order terms. The exponential-quadratic form is therefore canonical.

D Proof of Proposition 6 (Stochastic Limit Cycle)

This appendix provides the proof of Proposition 6 in the main text, which states that the spectral density of $n(t)$ under small stochastic forcing peaks at $\omega = \sqrt{\varphi/\mu}$.

Action–angle coordinates Write the deterministic solution as $n(t) = n_p + A\cos\phi(t)$ and $\dot{n}(t) = -A\omega\sin\phi(t)$, where $\phi(t) = \omega t + \phi_0$ and $\omega = \sqrt{\varphi/\mu}$. Transform the stochastic system to action–angle coordinates (A, ϕ).

Under the perturbation $\tilde{\pi} = \pi + \sigma_\varepsilon \varepsilon_t$, the energy $E = -\frac{\varphi}{2}A^2$ evolves as:

$$\mathrm{d}E = \frac{\sigma_\varepsilon \varepsilon_t}{\pi(n)} \dot{n}\,\mathrm{d}t.$$

Since $\dot{n} = -A\omega\sin\phi$ and $\pi(n) \approx \pi_0 e^E$ near the equilibrium orbit:

$$\mathrm{d}A \approx -\frac{\sigma_\varepsilon}{\varphi A \pi_0 e^E} A\omega\sin\phi\,\varepsilon_t\,\mathrm{d}t.$$

Averaging over the fast phase ϕ, the amplitude A evolves as a diffusion with drift coefficient determined by the noise coupling.

Spectral density The process $n(t)$ is a stochastically forced harmonic oscillator:

$$\ddot{n} + \omega^2(n - n_p) = \text{noise},$$

where the noise term inherits the i.i.d. structure of ε_t. The transfer function of this oscillator is:

$$H(\nu) = \frac{1}{\omega^2 - \nu^2 + i\eta\nu},$$

where η is an effective damping coefficient arising from the amplitude diffusion. The power spectral density is $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$, which peaks at $\nu = \omega = \sqrt{\varphi/\mu}$. Since S_ε is flat (white noise), the peak location is determined entirely by ω, not by σ_ε.

Properties

(i) The spectral peak is at $\omega = \sqrt{\varphi/\mu}$, as claimed.

(ii) The peak location depends on (φ, μ) only; the shock variance σ_ε^2 affects the peak *height* and *width*, not its location.

(iii) The persistence of the cycle is endogenous: even i.i.d. shocks produce a spectral peak because the transfer function $|H(\nu)|^2$ concentrates power near ω. ■

Role of shock structure Shock variance σ_ε^2 scales the peak height and width without shifting its location, since $|H(\nu)|^2$ peaks at $\nu = \omega$ regardless of the level of S_ε. With persistent shocks, $S_\varepsilon(\nu)$ is no longer flat, and the observed spectral density $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$ peaks where the product is maximized. Since $|H(\nu)|^2$ peaks at ω while a persistent process (e.g., AR(1)) tilts S_ε toward low frequencies, the observed peak can shift slightly below ω. The displacement is of order ρ/Q^2, where ρ is the shock autocorrelation and $Q = \omega/\eta$ is the quality factor of the transfer function. For sharp peaks ($Q \gg 1$), the displacement is negligible.

E Proof of Proposition 3 (General Hump-Shaped Profits)

Proof. Let $\pi(n)$ be a C^2 function with $\pi(n) > 0$ for n in a neighborhood of n_p, $\pi'(n_p) = 0$, and $\pi''(n_p) < 0$. The ironing condition gives:

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E.$$

Define the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which captures how the profit landscape confines the economy near optimal scale. Since $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, we have $U'(n_p) = 0$ and $U''(n_p) = -\pi''(n_p)/(\mu\pi(n_p)) > 0$, so n_p is a local minimum of U. This becomes:

$$\frac{1}{2}\dot{n}^2 + U(n) = -\frac{E}{\mu} \equiv \mathcal{E}.$$

Economically, this says that the sum of the profit-landscape cost $U(n)$ and the adjustment cost $\frac{1}{2}\dot{n}^2$ is constant along the equilibrium path. The economy oscillates around n_p because deviations from optimal scale are offset by adjustment dynamics. For $\mathcal{E}$ slightly above $U(n_p)$, the oscillations have small amplitude. The frequency of small oscillations is:

$$\omega = \sqrt{U''(n_p)} = \sqrt{\frac{-\pi''(n_p)}{\mu\,\pi(n_p)}} = \sqrt{\frac{|f'(n_p)|}{\mu}},$$

where $f'(n_p) = [\pi''/\pi - (\pi'/\pi)^2]|_{n_p} = \pi''(n_p)/\pi(n_p)$ since $\pi'(n_p) = 0$.

For finite amplitude A, the period of oscillation under a general profit landscape is:

$$T(A) = \sqrt{2} \int_{n_-}^{n_+} \frac{\mathrm{d}n}{\sqrt{\mathcal{E} - U(n)}},$$

where $n_\pm$ are the turning points. Expanding $U(n)$ to fourth order around n_p:

$$U(n) = U(n_p) + \frac{1}{2}U''(n_p)(n - n_p)^2 + \frac{1}{6}U'''(n_p)(n - n_p)^3 + \frac{1}{24}U^{(4)}(n_p)(n - n_p)^4 + \cdots$$

Standard perturbation theory for anharmonic oscillators gives:

$$\omega(A) = \omega_0 + O(A^2),$$

where $\omega_0 = \sqrt{U''(n_p)} = \sqrt{|f'(n_p)|/\mu}$.

For the Gaussian specification, $U(n) = \frac{\varphi}{2\mu}(n - n_p)^2 + \text{const}$ exactly (all higher-order terms vanish), so $\omega = \omega_0$ exactly for all A. For $\pi(n) = n^\alpha(a - bn)$, the cubic and quartic corrections are nonzero, producing amplitude-dependent corrections to the frequency. ■

F Entry at All Dates: Verification of the Regularity Condition

This appendix verifies that the regularity condition "entry occurs at all dates" in Proposition 2 is satisfied under the stated conditions.

Proposition 1.
Suppose $M > n_p + A$ (the total population exceeds the maximum of $n(t)$ along the cycle). Then in the timing search equilibrium of Theorem 1, the searcher pool satisfies $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t, and entry occurs at all dates.

Proof. Along the equilibrium path, $n(t) = n_p + A\cos(\omega t + \phi) \leq n_p + A < M$. Therefore $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t.

It remains to show that $e(t) > 0$ for all t. From the aggregate consistency condition, $e(t) = \dot{n}(t) + \delta n(t)$. Using $n(t) = n_p + A\cos(\omega t + \phi)$:

$$e(t) = -A\omega\sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

For $e(t) > 0$ to hold at all dates, we need $\delta n_p > A\sqrt{\omega^2 + \delta^2}$, i.e., the exit flow δn_p must exceed the maximum amplitude of the net flow $\dot{n} + \delta A \cos(\cdot)$. This is satisfied whenever the cycle amplitude is moderate relative to the steady-state scale: $A < \delta n_p / \sqrt{\omega^2 + \delta^2}$. ■

G Detailed Derivation of Entry-Matching Micro-Foundation

This appendix provides the detailed derivation of Proposition 1 in the main text (micro-foundation of the timing friction from entry-exit matching congestion).

Setup Under Assumption 3, the matching function is $m(S, R) = S^\alpha R^{1-\alpha}$, where $S = M - n$ is the searcher pool and R is the stock of entry resources. The matching probability per searcher is $p(\theta_e) = \theta_e^{-(1-\alpha)}$ with $\theta_e = S/R$. The matching infrastructure R is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$.

Steady state At $\dot{n} = 0$, the realized flow equals $e_{ss} = \delta n_p$, the infrastructure operates at its tuned capacity, and the matching probability equals $p_{ss} = p(\theta_e^{ss})$.

Bottleneck strain When $\dot{n} \neq 0$, the realized flow $|e - e_{ss}| = |\dot{n}|$ deviates from the tuned capacity in either direction. Rapid entry ($\dot{n} > 0$) strains the entry pipeline (permits, construction, specialist labor). Rapid net exit ($\dot{n} < 0$) strains the reallocation pipeline (resale markets, write-down procedures, workforce reassignment). Both reduce the matching probability per searcher below p_{ss}.

We represent this strain by a CES-like penalty that enters multiplicatively into the matching probability:

$$p(\theta_e; \dot{n}) = \theta_e^{-(1-\alpha)} \cdot \left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right]^{-(1-\alpha)/\alpha}. \tag{1}$$

The strain term depends on $\dot{n}^2$ rather than on signed $\dot{n}$, encoding the symmetry between entry and exit congestion. The $(\delta n_p)^2$ scale in the denominator reflects that infrastructure capacity is calibrated to steady-state throughput: deviations small

relative to δn_p are handled easily, while deviations approaching δn_p in magnitude strain the system to saturation.

What is assumed versus what is derived The specific CES functional form in (1) is a modeling choice: any smooth, symmetric strain penalty $g\big(\dot{n}^2/(\delta n_p)^2\big)$ with $g(0) = 0$ and $g'(0) = 1/2$ yields the same second-order behavior. The universality result in Online Appendix C shows that *any* smooth symmetric friction ξ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$ with $\mu = -\xi''(0)$; the second-order coefficient is therefore invariant to the particular smoothing chosen. What the specific form in (1) delivers is not the quadratic *shape* of the friction—that shape is universal—but the specific *scaling formula*

$$\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2},$$

which pins the constant in terms of the matching elasticity α and the steady-state throughput δn_p. This scaling follows from two observations that hold for any reasonable smoothing: (i) the only natural scale for $\dot{n}$ is δn_p (steady-state throughput), so μ must be inversely proportional to $(\delta n_p)^2$ by dimensional reasoning; (ii) the matching elasticity α enters through the Cobb-Douglas term $\theta_e^{-(1-\alpha)}$, producing the $(1-\alpha)/\alpha$ factor regardless of which specific strain function is chosen. The CES form is a convenient representative of this equivalence class; alternatives (e.g., Gaussian strain $\exp[-\dot{n}^2/(2(\delta n_p)^2)]$, or quadratic strain $[1 - \dot{n}^2/(2(\delta n_p)^2)]$) yield the identical μ at second order.

Effective friction The effective output friction from entry-exit congestion is:

$$\ln \xi(\dot{n}) \equiv \ln p(\theta_e;\, \dot{n}) - \ln p(\theta_e^{ss}) = -\frac{1-\alpha}{\alpha} \ln\left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right].$$

Taylor expansion For small $|\dot{n}|/(\delta n_p)$, $\ln(1+x) = x - x^2/2 + \cdots$, and only the leading term contributes at second order in $\dot{n}$:

$$\ln \xi(\dot{n}) \approx -\frac{1-\alpha}{\alpha} \cdot \frac{\dot{n}^2}{2(\delta n_p)^2} + O(\dot{n}^4).$$

This is $-\frac{\mu}{2}\dot{n}^2$ with

$$\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2} > 0.$$

Verification The friction ξ is symmetric ($\xi(\dot{n}) = \xi(-\dot{n})$), correctly signed ($\xi < 1$ for $\dot{n} \neq 0$), smooth (all derivatives exist at $\dot{n} = 0$), and matches the canonical quadratic form of Online Appendix C to leading order. For $\alpha = 0.5$ (symmetric matching), $\mu = 1/(\delta n_p)^2$. For $\alpha = 0.7$ (searcher-weighted), $\mu = 3/(7(\delta n_p)^2) \approx 0.43/(\delta n_p)^2$, approximately 57% smaller—implying faster cycles in industries with higher searcher matching power.

H Lindstedt-Poincaré Derivation of Frequency Correction

This appendix derives the $O(A^2)$ frequency correction for non-Gaussian profits stated in Proposition 4 of the main text.

Setup Consider the oscillator $\ddot{x} + U'(x) = 0$, where $x = n - n_p$ and $U(x) = U_0 + \frac{1}{2}\omega_0^2 x^2 + \frac{1}{6}\beta x^3 + \frac{1}{24}\gamma x^4 + \cdots$, with $\omega_0 = \sqrt{U''(0)}$, $\beta = U'''(0)$, $\gamma = U^{(4)}(0)$.

Lindstedt-Poincaré expansion. Seek a periodic solution:

$$x(t) = A\cos(\omega t) + A^2 x_2(t) + A^3 x_3(t) + \cdots ,$$

$$\omega = \omega_0 + A^2\omega_2 + O(A^4).$$

(Odd powers of A vanish by symmetry when $\beta = 0$.)

Order A^2 Substituting into the equation of motion and collecting $O(A^2)$ terms:

$$\ddot{x}_2 + \omega_0^2 x_2 = -\frac{\beta}{2}\cos^2(\omega_0 t) = -\frac{\beta}{4}[1 + \cos(2\omega_0 t)].$$

The particular solution is:

$$x_2(t) = -\frac{\beta}{4\omega_0^2} + \frac{\beta}{12\omega_0^2}\cos(2\omega_0 t).$$

Order A^3 (secular condition) At $O(A^3)$, the equation becomes:

$$\ddot{x}_3 + \omega_0^2 x_3 = \left[2\omega_0\omega_2 - \frac{\gamma}{8} + \frac{5\beta^2}{12\omega_0^2}\right]\cos(\omega_0 t) + \cdots$$

Eliminating the secular term (resonant forcing at frequency ω_0):

$$\omega_2 = \frac{1}{2\omega_0}\left[\frac{\gamma}{8} - \frac{5\beta^2}{12\omega_0^2}\right].$$

Result The corrected frequency is:

$$\omega(A) = \omega_0\left[1 + \left(\frac{\gamma}{16\omega_0^2} - \frac{5\beta^2}{12\omega_0^4}\right)A^2 + O(A^4)\right].$$

For the Gaussian profit: $U(x) = \frac{\varphi}{2\mu}x^2$ exactly (no higher terms), so $\beta = \gamma = 0$ and $\omega(A) = \omega_0$ for all A (exact isochrony).

For the quartic perturbation $\pi(n) = \pi_0 e^{-\frac{\varphi}{2}x^2 - a_4 x^4}$: $U(x) = \frac{\varphi}{2\mu}x^2 + \frac{a_4}{\mu}x^4$, so $\beta = 0$ and $\gamma = 24a_4/\mu$. Then $\omega_2 = \frac{3a_4}{\mu\omega_0} = 3a_4/\varphi$ (using $\omega_0^2 = \varphi/\mu$), giving:

$$\omega(A) = \omega_0\left(1 + \frac{3a_4}{\varphi}A^2\right),$$

confirming the Duffing-oscillator formula used in the numerical experiments.

Cycle asymmetry (Proposition 5) When $\beta = U'''(0) \neq 0$, the $O(A^2)$ correction $x_2(t)$ includes a constant shift $-\beta/(4\omega_0^2)$ and a second-harmonic term. The constant shift displaces the time-averaged position away from n_p, making the orbit spend unequal time above and below n_p. The leading-order asymmetry parameter is:

$$\mathcal{A} \equiv \frac{t_{\text{above}} - t_{\text{below}}}{t_{\text{below}}} = \frac{\beta}{4\omega_0^2}A + O(A^2).$$

Expressing β through the profit function yields $U'''(0) = -[\pi'''(n_p)\pi(n_p) - 3\pi''(n_p)\pi'(n_p)]/(\mu\pi(n_p)^2)$. Since $\pi'(n_p) = 0$, this simplifies to $\beta = -\pi'''(n_p)/(\mu\pi(n_p))$. Using $\omega_0^2 = |\pi''(n_p)|/(\mu\pi(n_p))$:

$$\mathcal{A} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}A + O(A^2).$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), $\mathcal{A} < 0$: the economy spends more time below n_p, producing slow expansions and sharp contractions. ■

I Stationary Distribution of the Damped Stochastic Oscillator

This appendix first derives the damped stochastic oscillator directly from the ironing condition with entry-cost dissipation, then derives the stationary amplitude distribution stated in Proposition 7 of the main text. The derivation uses an energy-balance argument that avoids any action-principle ambiguity.

Conservative equation from ironing Under Gaussian profits (Assumption 2) and Gaussian friction (Proposition 1), the ironing condition (Proposition 2 of the main text) states $\pi(n)\xi(\dot{n}) = C$ along any equilibrium path. Linearizing around $(n_p, 0)$ and writing $x = n - n_p$:

$$\ln \pi(n_p) - \frac{\varphi}{2}x^2 - \frac{\mu}{2}\dot{n}^2 = \ln C.$$

Differentiating in time and cancelling the non-trivial factor $\dot{n} \neq 0$:

$$-\varphi\, x - \mu\, \ddot{n} = 0 \quad \Longleftrightarrow \quad \mu\, \ddot{n} + \varphi\, x = 0,$$

which is Newton's equation for a harmonic oscillator with frequency $\omega_0 = \sqrt{\varphi/\mu}$. This reproduces Theorem 1: the Hamiltonian

$$\mathcal{H} \equiv \tfrac{1}{2}\,\mu\,\dot{n}^2 + \tfrac{1}{2}\,\varphi\, x^2$$

is constant along the undamped equilibrium path, $\mathrm{d}\mathcal{H}/\mathrm{d}t = 0$.

Dissipation from convex entry-exit congestion The entry-exit process consumes real resources: permits, construction labor, site preparation for entrants; reorganization services, write-downs, workforce reassignment for exiters. Let $c(\dot{n})$ denote the industry's real resource cost per unit time, measured in output units. The function c is smooth and convex around $\dot{n} = 0$ (steady-state flow), so it admits the Taylor expansion

$$c(\dot{n}) = c(0) + c'(0)\,\dot{n} + \tfrac{1}{2}\,c''(0)\,\dot{n}^2 + O(\dot{n}^3).$$

Call $\kappa \equiv c''(0) > 0$ the convex congestion coefficient. The zeroth and first terms, $c(0)$ and $c'(0)\dot{n}$, represent a constant baseline cost and a cost linear in the flow. Integrated over a cycle, the linear term averages to zero (no net energy loss) and the constant simply scales C; neither dissipates energy. The second-order term, by contrast, represents a strictly positive rate of output loss whenever $\dot{n} \neq 0$:

$$\text{rate of energy dissipation} = \kappa\,\dot{n}^2 \geq 0.$$

Only the convex part of the entry-cost function dissipates energy; this is a direct consequence of convexity (Jensen's inequality applied to the cost per unit rate).

Energy-balance derivation of the damped equation Along any equilibrium path with entry-cost dissipation, the Hamiltonian $\mathcal{H}$ falls at exactly the rate at which the bottleneck consumes output:

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = -\kappa\,\dot{n}^2.$$

Computing the time derivative of $\mathcal{H}$ directly:

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \mu\,\dot{n}\,\ddot{n} + \varphi\,x\,\dot{n} = \dot{n}\left(\mu\,\ddot{n} + \varphi\,x\right).$$

Equating and cancelling $\dot{n}$ (for $\dot{n} \neq 0$):

$$\mu\,\ddot{n} + \varphi\,x = -\kappa\,\dot{n}.$$

Rearranging:

$$\mu\,\ddot{n} + \kappa\,\dot{n} + \varphi\,x = 0.$$

Dividing by μ and adding the stochastic forcing $\sigma_\varepsilon \varepsilon_t/\mu$ from Section 4 of the main text gives the damped stochastic oscillator

$$\ddot{n} + \eta\,\dot{n} + \omega_0^2\,x = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t,$$

with $\omega_0 = \sqrt{\varphi/\mu}$ and

$$\boxed{\eta = \frac{\kappa}{\mu}}.$$

Both signs are correct by construction: the restoring force $-\varphi x$ pulls the industry toward n_p, and the dissipative force $-\kappa \dot{n}$ opposes motion. The damping rate η emerges endogenously from the ratio of the convex entry-cost coefficient κ (the rate at which the bottleneck consumes output) to the matching friction μ (how stiffly the ironing condition pins the motion). Higher congestion cost dissipates energy faster; higher matching friction slows both the restoring force and the dissipation by the same factor.

Remark on κ The coefficient $\kappa = c''(0)$ is the convex part of the entry-processing cost. This parameter overloads the per-firm entry cost appearing in the free-entry condition $V(t) = V^{\text{search}}(t) + \kappa$ (Definition 1(ii) of the main text): both scale with the intensity of the entry-matching bottleneck, so that a policy intervention that lowers the per-firm cost (e.g., entry subsidies) also lowers the convex coefficient proportionally. The two are structurally linked by the shared infrastructure from which they arise; for clean derivations it is sufficient to track one parameter.

Setup With the damped stochastic oscillator thus derived, the remainder of this appendix computes its stationary amplitude distribution.

Energy equation Define the energy $\mathcal{E} = \frac{1}{2}\dot{n}^2 + \frac{1}{2}\omega_0^2(n - n_p)^2$. Then:

$$\mathrm{d}\mathcal{E} = -\eta\, \dot{n}^2\, \mathrm{d}t + \frac{\sigma_\varepsilon}{\mu}\, \dot{n}\, \mathrm{d}W_t + \frac{\sigma_\varepsilon^2}{2\mu^2}\, \mathrm{d}t.$$

Stationarity condition In stationarity, $\mathbb{E}[\mathrm{d}\mathcal{E}] = 0$:

$$\eta\, \mathbb{E}[\dot{n}^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Equipartition For the linear damped oscillator with white noise, the stationary distribution of $(n - n_p, \dot{n})$ is bivariate Gaussian. Equipartition gives:

$$\mathbb{E}[\dot{n}^2] = \omega_0^2\, \mathbb{E}[(n - n_p)^2].$$

The Hilbert-envelope amplitude satisfies $\mathbb{E}[A^2] = 2\mathbb{E}[(n - n_p)^2]$. Therefore:

$$\eta \cdot \frac{1}{2}\omega_0^2\, \mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Solving:

$$\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{\mu^2\,\eta\,\omega_0^2} = \frac{\sigma_\varepsilon^2}{\mu\,\eta\,\varphi}.$$

Rayleigh distribution The pair $(n - n_p, \dot{n}/\omega_0)$ is bivariate Gaussian with equal variances $\sigma_x^2 = \sigma_\varepsilon^2/(2\mu\eta\varphi)$. The envelope $A = \sqrt{(n-n_p)^2 + (\dot{n}/\omega_0)^2}$ therefore follows the Rayleigh distribution:

$$f(A) = \frac{A}{\sigma_x^2}\exp\left(-\frac{A^2}{2\sigma_x^2}\right), \qquad A \geq 0.$$

The mode is at σ_x and the mean is $\sigma_x\sqrt{\pi/2}$. Both scale linearly with σ_ε, confirming the linear relationship $\sigma_x \propto \sigma_\varepsilon$ (and equivalently $\sqrt{\mathbb{E}[A^2]} \propto \sigma_\varepsilon$). ■

J Equilibrium Verification

This appendix verifies that the path constructed in Theorem 1 satisfies all five conditions of the timing search equilibrium (Definition 1).

Entry policy and aggregate consistency (iv) The entry policy implied by the equilibrium path is $e(t) = \dot{n}(t) + \delta n(t)$. Using $n(t) = n_p + A\cos(\omega t + \phi)$:

$$e(t) = -A\omega\sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

This satisfies $\dot{n} = e - \delta n$ by construction.

Entry non-negativity For $e(t) \geq 0$ at all dates, we need $\delta n_p > A\sqrt{\omega^2 + \delta^2}$, i.e., the steady-state exit flow must exceed the maximum amplitude of the net flow. This holds whenever the cycle amplitude is moderate relative to steady-state scale: $A < \delta n_p/\sqrt{\omega^2 + \delta^2}$.

Sufficiency of the ironing condition (i, iii) The ironing condition gives $\pi(n(t))\xi(\dot{n}(t)) = C$ for all t. Substituting this constant into the value function $V(\tau) = \int_\tau^\infty e^{-(r+\delta)(t-\tau)}\pi(n(t))\xi(\dot{n}(t))\,\mathrm{d}t$ from the main text yields

$$V(\tau) = \int_\tau^\infty e^{-(r+\delta)(t-\tau)}\,C\,\mathrm{d}t = \frac{C}{r+\delta} \equiv V^*$$

for all τ. Since $V(\tau)$ is the same at every date, each agent's entry date is optimal (condition (i)) and no agent can gain by retiming (condition (iii)). The ironing condition is therefore sufficient, not just necessary, for individual optimality.

Temporal free entry (ii) Along the equilibrium path, $V(\tau) = V^* = C/(r+\delta)$ for all τ. Free entry requires $V(\tau) = V^{\text{search}}(\tau) + \kappa$, which pins $V^{\text{search}} = V^* - \kappa$ at all dates. Since V^* is constant, this is satisfied identically.

Feasibility (v) The maximum of $n(t)$ is $n_p + A$. Feasibility requires $n_p + A \leq M$, i.e., the total population must exceed the peak number of active firms. With A moderate and M large, this holds.

K Additional Simulation Results

This appendix provides additional computational verification of the timing search model, complementing the spectral density comparison in the main text. Parameters: $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, $n_p = 1$ (theoretical prediction: $\omega^* = 2$, $T^* = \pi \approx 3.14$ time units).

Phase portrait The deterministic equilibrium orbit in the $(n - n_p, \dot{n})$ phase plane is a closed ellipse traversed clockwise, confirming the exact harmonic oscillation of Theorem 1. Under stochastic forcing, orbits remain approximately elliptical but migrate across amplitudes as E drifts under shocks, with $\mathrm{d}E \approx (\sigma_\varepsilon/\pi(n))\dot{n}\,\mathrm{d}W$. This amplitude diffusion produces variation in cycle severity without altering cycle frequency—the stochastic analog of isochrony (Remark 1 of the main text).

BDS test for nonlinear dependence The BDS test (Brock, Dechert, and Scheinkman, 1996) examines whether AR(1)-filtered residuals contain nonlinear serial dependence. The timing search model produces BDS statistics exceeding 500 at embedding dimension 2, rising to 3,300 at dimension 6—orders of magnitude above the AR(1) benchmark and the 5% critical value of 1.96. The escalation with embedding dimension reflects the deterministic skeleton underlying the stochastic dynamics: the ironing condition confines the economy to a two-dimensional manifold in $(n, \dot{n})$ space,

and higher-dimensional embeddings detect this geometric structure with increasing power.